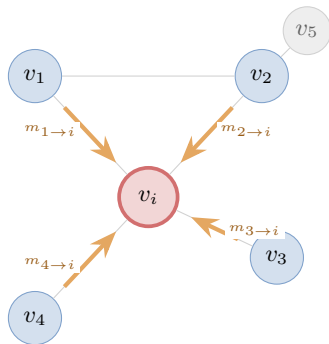
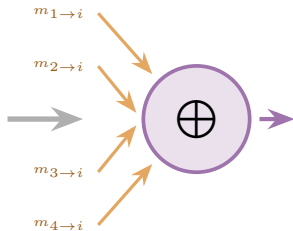


1. Mensaje



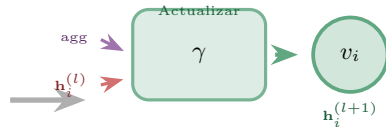
$$m_{j \rightarrow i} = \phi(\mathbf{h}_j^{(l)}, \mathbf{h}_i^{(l)})$$

2. Agregación



$$\bigoplus_{j \in \mathcal{N}(i)} m_{j \rightarrow i}$$

3. Actualización



$$\mathbf{h}_i^{(l+1)} = \gamma(\mathbf{h}_i^{(l)}, \bigoplus_{j \in \mathcal{N}(i)} m_{j \rightarrow i})$$

Equivalencia con convolución en grafos: $(f \star K)(v_i) = \sum_{j \in \mathcal{N}(i)} K(v_j) f(v_j) \longleftrightarrow \mathbf{h}_i^{(l+1)} = \gamma(\mathbf{h}_i^{(l)}, \bigoplus_{j \in \mathcal{N}(i)} \phi(\mathbf{h}_j^{(l)}, \mathbf{h}_i^{(l)}))$